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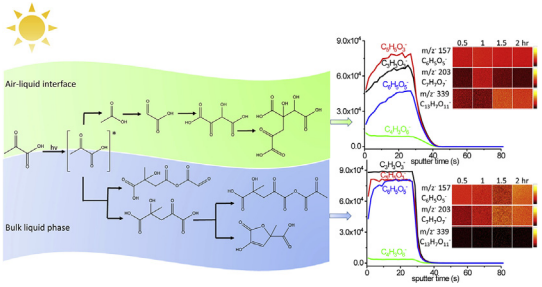
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Does interfacial photochemistry play a role in the photolysis of pyruvic acid in water?

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GRAPHICAL ABSTRACT



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ABSTRACT

Pyruvic acid (PA) exists in fogs, aerosols and clouds. The photochemistry-driven reaction pathways of PA in the aqueous phase are more elusive than the gas phase. The PA photochemical process may occur in the bulk liquid phase and at the air-liquid interface in ambient conditions. We conducted two sample preparation methods to simulate two possible scenarios: the air-liquid interface and the bulk liquid phase under photolysis. Time-of-flight secondary ion mass spectrometer (ToF-SIMS) was used to analyze samples because of its high sensitivity and mass accuracy in surface analysis. Both negative and positive ion mode mass spectra provide complementary information of the products under different reaction conditions. Spectral principal component analysis (PCA) is used to determine similarities and differences among various samples. The air-liquid interface facilitates more radical reactions and form higher molecular weight compounds (HMWCs) more quickly than the bulk liquid phase, which mainly has non-radical reactions such as anhydride reactions and decarboxylation reactions. Our results show that interfacial chemistry plays an important role in atmospheric scenarios. Moreover, different types of secondary organic aerosols (SOAs) are formed, suggesting the strong influence of interfacial photochemistry has on the earth atmosphere.

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1. Introduction

The air-liquid interface is inherently an asymmetric environment (Pruyne et al., 2014), where large changes over angstrom-scale distances in properties such as molecular composition, orientation, and density take place, leading to differences in the structure and chemistry between the interface and the bulk phase (Eisenthal, 1993; Petersen and Saykally, 2006). For example, the pH at the interface can be either more acidic or alkaline than that in the bulk phase (Buch et al., 2007; Petersen et al., 2004). Due to highly varied concentrations in the aqueous organic film at the air-liquid interface under solar radiation, photochemistry is known to occur at the aerosol surface (Donaldson and Vaida, 2006).

Interfacial processes have become a noticeable topic in atmospheric chemistry (George et al., 2015) due to its perceived importance in secondary organic aerosol (SOA) formation. Recent findings have shown that the interface, where specific chemical and sunlight-initiated photochemical reactions take place, can produce high-molecular-weight compounds (HMWCs) and aggregates (Baergen and Donaldson, 2013; Brüggemann et al., 2017; Ciuraru et al., 2015; Dupart et al., 2012; Enami et al., 2015; Fu et al., 2015; Griffith et al., 2014; Reed Harris et al., 2014; Renard et al., 2014; Rossignol et al., 2016). These products will affect SOA formation and organic mass, chemical composition, and optical properties of SOAs, resulting in a significant impact on climate, air quality, and human health (Vaida, 2016).

As a key oxidation product of isoprene in the atmosphere (Carlton et al., 2006; Ervens et al., 2011), pyruvic acid (PA) is observed in both gas and aqueous phases (Altieri et al., 2006; Andreae et al., 1987; Talbot et al., 1990; Veres et al., 2010; Warneck, 2005). PA is the simplest α-keto acid in the atmosphere, and it can absorb light in the near-UV region from the solar photon flux (Horowitz et al., 2001). The reactions of PA occur in the gas phase and in the aqueous phase, with photolysis or dark aging in a range of reactant concentrations and relative humidity (Altieri et al., 2006; Chang et al., 2014; Guzman et al., 2006; Larsen and Vaida, 2012; Lopalco et al., 2016; Mellouki and Mu, 2003; Plath et al., 2009; Rapf et al., 2017; Reed Harris et al., 2014, 2016; Vesley and Leermakers, 1964).

The photolysis rate of PA is faster than other reactions, and photochemistry is the main atmospheric sink of PA (Lopalco et al., 2016; Mellouki and Mu, 2003; Reed Harris et al., 2014; Schöne and Herrmann, 2014). For instance, PA can release CO_2 and acetaldehyde due to photolysis according to earlier works (Chang et al., 2014; Mellouki and Mu, 2003; Takahashi et al., 2008). The photochemical pathways of PA are still under debate. Especially in the aqueous phase, the photochemistry is considered different due to the water environment (Vaida, 2011). For example, the electronic structure of PA will change and generate organic radicals (e.g., $CH_3\dot{C}(OH)$ CO_2H, $\dot{C}H_2C(O)CO_2H$) in water (Rapf et al., 2017). These radicals will then recombine to form oligomers, contributing to the production of SOAs (Boris et al., 2014; Eugene and Guzman, 2017; Griffith et al., 2013; Guzman et al., 2006; Leermakers and Vesley, 1963; Lim et al., 2010; Reed Harris et al., 2014). Although SOA is perceived as a main contributor of interfacial reactions in atmospheric aerosols and cloud water, the challenge to clearly obtain organic speciation often existed in trace amount remains one of the grand challenges in atmospheric chemistry (Carlton et al., 2006, 2009; Ervens et al., 2011). Therefore, it is of significant importance to analyze the surface composition because it can differ from that in the bulk (Jungwirth and Tobias, 2006; Krieger et al., 2012; Shiraiwa et al., 2010).

Among surface analytical techniques, time-of-flight secondary ion mass spectrometry (ToF-SIMS) is a powerful tool and has been widely used in many research areas such as contamination, adhesion, corrosion, cell chemistry, and biocompatibility (Fearn, 2015; Hagenhoff, 2000). SIMS is a highly sensitive surface analysis technique, and it can provide detailed elemental and molecular mapping of solid surfaces, thin films, gas-solid interfaces, and three dimensional chemical mapping of solid samples (Benninghoven, 1994; Lanni et al., 2014; Robinson et al., 2012; Yu et al., 2010). Previous studies have shown that SIMS was a useful technique for exploring aerosol surface characterization. Not only inorganic composition but also organic compounds including hydrocarbons, alcohols, aldehydes, ethers, carboxylic acids, amines, amides, nitriles and a number of polycyclic aromatic hydrocarbons (PAHs) and nitroPAHs can be studied using SIMS (Chan et al., 2018; Huang et al., 2017; Peterson and Tyler, 2002, 2003; Tervahattu, 2002; Zhang et al., 2016; Zhu et al., 2001).

In this work, PA photochemistry occurred at the air-liquid interface and bulk liquid phase were studied using ToF-SIMS. Different dry sample preparation steps were used to simulate two possible atmospheric reaction scenarios: 1) the bulk liquid phase and 2) the air-liquid interface. The photochemical process occurred at the air-liquid interface could lead to aqueous SOA (aqSOA) formation (Bernard et al., 2016; Brüggemann et al., 2017). It is postulated that distinct photochemical processes happened at the air-liquid interface and in the bulk phase, which lead to the formation of dissimilar products. HMWCs, a main constituent of SOAs, are discovered to form at the air-liquid interface due to more photochemical oxidation compared to the bulk liquid photolysis as summarized in Fig. 1.

2. Experimental information and methods

2.1. Chemicals

Pyruvic acid (98%, Sigma-Aldrich, St. Louis, MO, USA) was diluted with DI water (18.2 MΩ) dispensed from a Barnstead water purification system (model: Nanopure diamond) to prepare a 0.1 M solution. The initial pH of the 0.1 M PA solution was ~1.7 measured by a portable pH meter (accumet, Fisher Scientific, CA, USA).

2.2. Photolysis and sample substrates

A Hg-Ar UV lamp (ORIEL, lamp model 6035, USA) with a power supply (ORIEL model 6060, USA) was used as the UV source for photochemical aging, and the wavelength was 334.15 nm. Si wafers (10 × 10 mm diced, Ted Pella Inc., CA, USA) were cleaned by acetone, isopropanol, and high-purity DI water for 3 min sequentially. After thorough cleaning, UV-O_3 plasma (Model No.342, Jelight Company Inc., USA) was applied to remove residual organic contaminations on the Si wafer surface for 2 min. Afterwards, the wafers were used as substrates for dry sample preparation (Jensen et al., 2013; Kanyal et al., 2015; Mahoney et al., 2006; Song et al., 2011; Sui et al., 2018). The distance between the Si wafer surface and the UV lamp was set at 10 cm (Sui et al., 2018). All sample preparation was conducted in a fume hood (Hamilton Laboratory Solutions LLC., WI, USA). The samples were analyzed immediately after photolysis treatment to reduce possible interferences.

2.3. Sample preparation

The details of samples studied in this work are summarized in Fig. S1 and Table S1. In order to simulate the bulk and the air-liquid interface scenarios, two methods were performed to prepare dry samples in Fig. 2. The first method (Fig. 2a) was described in the following steps: (1) the Si wafer was put in a clean petri dish; (2) then 20 μL of sample solution was taken and deposited it on the cleaned Si wafer; and (3) the droplet was dried under the UV light for a specific time (0.5 h, 1 h, 1.5 h, and 2 h). This method simulated the photochemistry at the air-liquid interface. The operation sequence of the second preparation approach (Fig. 2b) was different from those in the first one: (1) the sample solution of 10 mL was stored in a glass vessel under the UV light for a specific time (0.5 h, 1 h, 1.5 h, and 2 h); (2) 20 μL of solution was taken out of the vessel immediately after the photochemical aging was ended; (3) the solution was deposited on a treated Si wafer in a clean

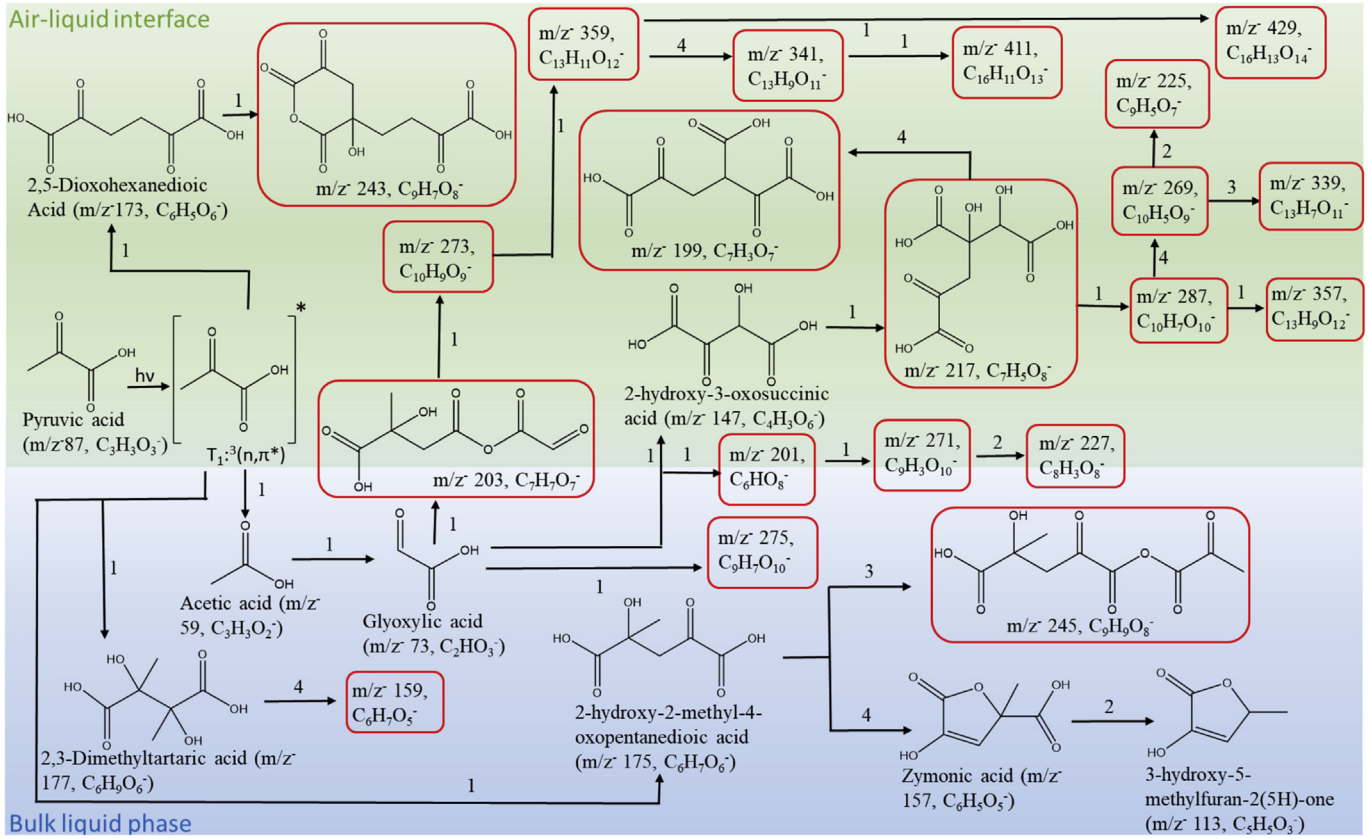


Fig. 1. A reaction flow chart showing reactions in the bulk liquid phase and the air-liquid interface. The numbers on the arrow indicate different reaction schemes, e.g., Reaction (Rxn.) 1 represents radical reactions (Rapf et al., 2017), Rxn. 2 decarboxylation reactions (Tan et al., 2012), Rxn. 3 anhydride formation and Rxn. 4 esterification formation (Lim et al., 2010). Red squares indicate new products observed in this study. When the structure is too big or undetermined, only the chemical formula are presented. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

petri dish and covered with a lid; and (4) the droplet was dried in air in a fume hood. The second method was intended to simulate the photochemistry occurred in the bulk liquid phase. Additionally, a system control (Fig. 2c) was prepared to characterize the sample matrix. In this case, a drop of the 0.1 M PA solution was directly deposited on a clean Si wafer, covered by a lid and, then dried in air without UV illumination in the fume hood. In the drying process, a sample was typically thoroughly dried at about 1.5 h for all cases.

Before being transferred into the main chamber of the ToF-SIMS instrument, half an hour of dark reaction time was inevitable for the photochemical aging samples due to vacuum pumping in the load lock chamber. Additional experimental details are shown in SI materials.

2.4. ToF-SIMS analysis

2.4.1. ToF-SIMS analysis of dry samples

Measurements were conducted in a ToF-SIMS V instrument (IONTOF GmbH, Münster, Germany). The pressure of the main chamber remained $\sim 1 \times 10^{-8}$ mbar during analysis. During spectral measurements, a pulsed 25 keV Bi_3^+ primary ion beam (10 kHz, pulse width 0.8 ns, ~0.64 pAcurrent) was used for scanning on an area of $200 \times 200\ \mu m^2$ on the dry sample surface. At least 4 positive and 4 negative data points were obtained at different locations of each sample. Data was collected for 100 scans (128 × 128 shots per scan) for each measurement. Figs. S2a and S2b show the data reproducibility of SIMS spectra in the negative and positive mode, respectively. In depth profiling, a 10 keV Argon cluster with a 2 nA beam current was used as the sputter beam with a crater size of $500 \times 500\ \mu m^2$. The analysis area was $200 \times 200\ \mu m^2$, and a flood gun with a current of 2.2 μA was used during depth profiling for charge compensation. Depth profiles (Fig. 7 and Fig. S9) were obtained in the positive and negative mode to study the compositional differences between the surface and the bulk of each sample.

2.4.2. Data analysis

Data was analyzed using the IONTOF Surface Lab 6.3 software. The mass spectra were calibrated by using H^+ (m/z^+ 1.01), CH_3^+ (m/z^+ 15.02), $C_2H_3^+$ (m/z^+ 27.02), $C_2H_5O^+$ (m/z^+ 45.03), $C_4H_7^+$ (m/z^+ 55.05) and $C_4H_5O_2^+$ (m/z^+ 85.03) in the positive mode and H^- (m/z^- 1.01), CH^- (m/z^- 13.01), CH_2^- (m/z^- 14.02), O^- (m/z^- 15.99), $C_2H_3O^-$ (m/z^- 43.02), $C_6H_5O_5^-$ (m/z^- 157.01) and $C_6H_7O_6^-$ (m/z^- 175.02) in the negative mode, respectively. For the high-resolution spectra results, the mass resolution was about 4000–7000. After mass calibration, data were exported and plotted by Origin Pro (2017). It is worth noting that SIMS was a semi-quantitative instrument, therefore the compounds discussed throughout the text were displayed with normalized intensity. To evaluate the intrinsic similarities and discriminations among samples, spectral principal component analysis (PCA) was conducted using Matlab (version R2013a). SIMS data were treated by the following steps: normalization to the total ion intensity of selected peaks, square root transformation, and mean-centering, prior to spectral PCA (Sui et al., 2017) after removing known interference peaks in the first round of analysis followed with selected peak spectral PCA. In the second round of spectral PCA, peaks were chosen based on the following criteria: (a) the loading of a peak was significantly higher than its neighboring peaks, (b) the intensity of the peak in the SIMS mass spectrum was 3 times higher than the its neighboring peaks, (c) the peak intensity was more than 0.5% of the total ion intensities, and (d) product peaks identified in previous papers were included in this study (Ding et al., 2016; Sui et al., 2017).

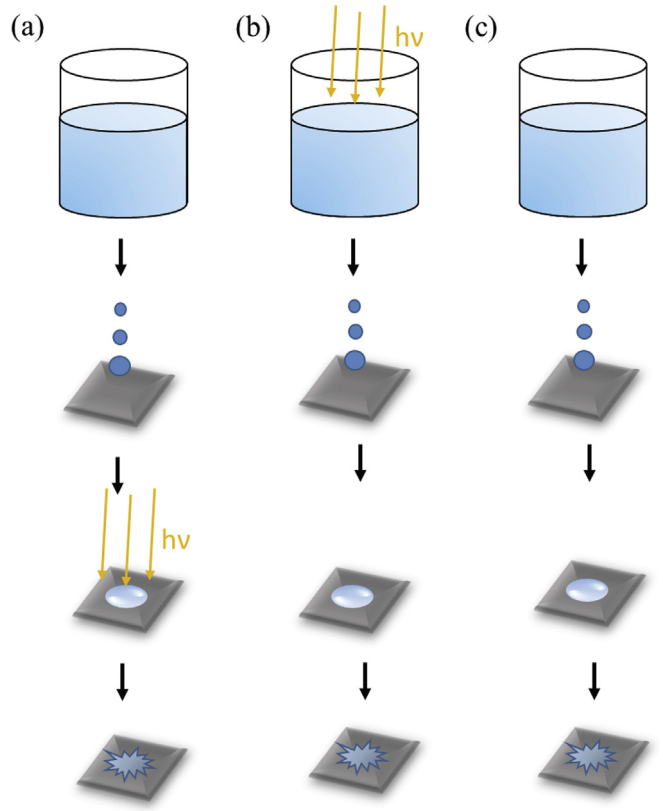


Fig. 2. A schematic showing sample preparation processes to simulate: (a) the air-liquid interface under photolysis; (b) the bulk liquid phase under photolysis; and (c) system control.

3. Results and discussions

3.1. SIMS spectral comparison of the air-liquid interface and the bulk liquid phase samples

Fig. 3 and Fig. 4 show the comparison of representative SIMS spectra results (m/z$^-$ 70–400) among samples after photolysis for different UV aging time at the air-liquid interface and in the bulk liquid phase in the negative ion mode. Additional spectra are shown in Figs. S4a- S5d. Each spectrum is selected randomly from four independent measurements. All the spectra are obtained without sputtering, which reflect the molecular information at the outermost surface of the sample. Fragments from the substrates (e.g., m/z$^-$ 60.97 SiO_2H^-, *m/z*$^+$ 44.98 $SiHO^+$) are observed due to the high sensitivity of SIMS. Tables S2 and S3 summarize possible interference peaks in the positive mode and negative modes, respectively (Kanyal et al., 2015).

The molecular peak of PA (m/z$^-$ 87.01, $C_3H_3O_3^-$) and some reported product peaks such as 3-hydroxy-5-methylfuran-2(5H)-one (m/z$^-$ 113.02, $C_5H_5O_3^-$), zymonic acid (m/z$^-$ 157.01, $C_6H_5O_5^-$), parapyruvic acid (m/z$^-$ 175.02, $C_6H_7O_6^-$), 2,3-dimethyltartaric acid (m/z$^-$ 177.03, $C_6H_9O_6^-$) are observed in both scenarios (Figs. 3 and 4) (Rapf et al., 2017). Additionally, other products with higher ion intensities are found to form in this work, such as $C_4H_3O_6^-$ (m/z$^-$ 146.99), $C_7H_3O_7^-$ (m/z$^-$ 198.98), $C_{10}H_5O_9^-$ (m/z$^-$ 268.98), $C_9H_3O_{10}^-$ (m/z$^-$ 270.97), $C_{13}H_9O_{11}^-$ (m/z$^-$ 341.03) and $C_{13}H_{11}O_{12}^-$ (m/z$^-$ 359.04) at the air-liquid interface (Fig. 3), and $C_5H_5O_4^-$ (m/z$^-$ 129.01), $C_7H_7O_7^-$ (m/z$^-$ 203.02), $C_{10}H_{11}O_9^-$ (m/z$^-$ 275.00), $C_{12}H_{15}O_9^-$ (m/z$^-$ 303.07), $C_{13}H_{15}O_{10}^-$ (m/z$^-$ 331.07), $C_{13}H_{15}O_{11}^-$ (m/z$^-$ 347.06) in the bulk liquid phase (Fig. 4). These results indicate the oxidation products vary due to different sample preparation methods. Moreover, as the photochemical aging time increases, HMWCs, m/z$^-$ from 200 to m/z$^-$ 550, begin to form both at the air-liquid interface and in the bulk liquid phase (Figs. 3 and 4, Figs. S4a and S4b), which suggests that both scenarios facilitate the formation of species with large molecular weight due to longer-time photochemical oxidation. A similar trend is found in the positive spectral comparison (Figs. S5a-d). Besides, both negative spectra and positive spectra demonstrate that the air-liquid interface samples show larger *m/z* peaks than the bulk liquid phase samples. This finding suggests that the air-liquid interfacial reactions lead to more higher mass products.

3.2. Key product identification

Possible peak identification of key reactant and products are listed in Tables 1 and S4a – S4c. The criteria of peak identification are described in the SI. Complex reactions occur in both scenarios producing

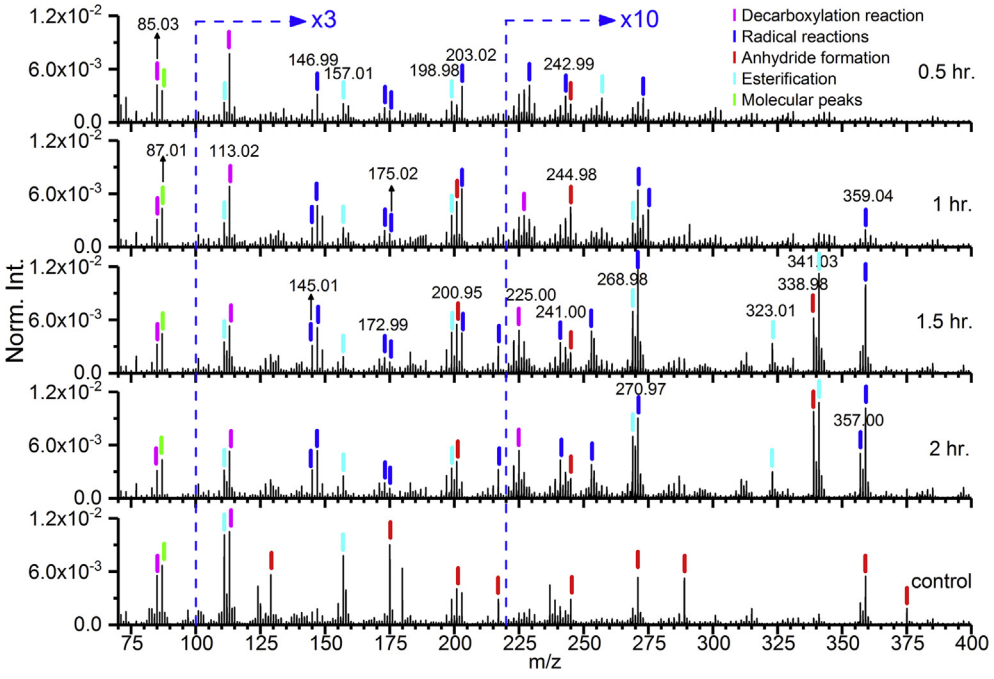


Fig. 3. Normalized negative SIMS spectral comparison of the air-liquid interface samples treated with UV photolysis ranging from 0.5 h to 2 h and the PA system control. Color bars represent products related to different reaction mechanisms. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

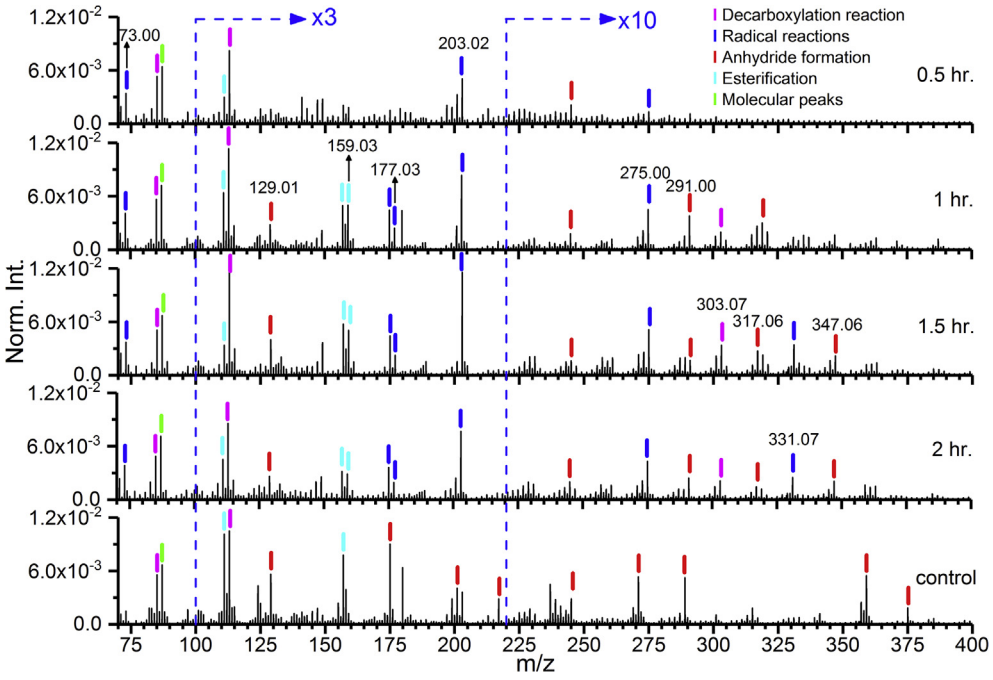


Fig. 4. Normalized negative SIMS spectral comparison of the bulk liquid phase samples treated with UV photolysis ranging from 0.5 h to 2 h and the PA system control. Color bars represent products related to different reaction mechanisms. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Possible peak identifications of key reactant and products in the negative SIMS spectra.

m/z^-_{obv} [a]	m/z^-_{theo} [b]	Formula	Peak assignment	References
72.996	72.993	$C_2HO_3^-$	Glyoxylic acid	(Tan et al., 2012)
85.033	85.029	$C_4H_5O_2^-$	Biacetyl	(Tan et al., 2012)
113.02	113.024	$C_5H_5O_3^-$	3-hydroxy-5-methylfuran-2(5H)-one	(Rapf et al., 2017)
157.012	157.014	$C_6H_5O_5^-$	Zymonic acid	(Perkins et al., 2016)
159.026	59.029	$C_6H_7O_5^-$	$C_6H_9O_6^- - H_2O$	This work
175.019	175.024	$C_6H_7O_6^-$	Parapyruvic acid	(Rapf et al., 2017)
203.015	203.019	$C_7H_7O_7^-$	$C_2H_3O^-$ ···m/z^- 73 − H_2O ··· $C_3H_5O_3$•	This work
146.987	146.993	$C_4H_3O_6^-$	1-hydroxy-2-oxoethyl hydrogen carbonate (C_2HO_3· ··· $C_2H_3O_3$·)	(Lim et al., 2010)
198.981	198.988	$C_7H_3O_7^-$	m/z^- 217 − H_2O	This work
216.993	216.998	$C_7H_5O_8^-$	$C_4H_5O_6$· ··· $C_3H_3O_3$· − H_2O	This work
224.996	225.003	$C_9H_5O_7^-$	m/z^- 269 − CO_2	This work
268.982	268.993	$C_{10}H_5O_9^-$	$C_{10}H_7O_{10}^- - H_2O$	This work
338.988	338.999	$C_{13}H_7O_{11}^-$	m/z^- 269 ··· m/z^- 87 − H_2O	This work
359.036	359.025	$C_{13}H_{11}O_{12}^-$	$C_{10}H_9O_9$· ··· $C_3H_3O_3$·/m/z^- 175 ···$C_6HO_8^-$ − H_2O	This work

[a] m/z^-_{obv}: the observed mass.
[b] m/z^-_{theo}: the theoretical mass.

various components labeled with color bars in Figs. 3 and 4. They are discussed in detail in this section. (see Table 1).

3.2.1. Radical reactions

PA can easily undergo photolysis to form its electronic excitation state (PA*) (Arnett et al., 1974). In the presence of PA and PA*, $CH_3\dot{C}(OH)CO_2H$, and $\dot{C}H_2C(O)CO_2H$ radicals are formed irreversibly by dehydrogenize from the methyl functional groups in the pyruvic acid molecules. These two radicals can form parapyruvic acid (m/z^- 175.02, $C_6H_7O_6^-$) in the acidic environment. 2,3-dimethyltartaric acid (m/z^- 177.03, $C_6H_9O_6^-$) is formed by combining two $CH_3C(OH)COOH$· radicals. Interestingly, these two acids (m/z^- 175.02 and m/z^- 177.03) have higher ion intensities in the bulk liquid phase than those at the air-liquid interface. We postulate that parapyruvic acid and 2,3-dimethyltartaric acid could immediately decompose to m/z^- 113.02 ($C_5H_5O_3^-$) by decarboxylation at the interface driven by photochemistry and water evaporation.

At the air-liquid interface, more compounds such as m/z^- 172.99 ($C_6H_5O_6^-$), m/z^- 203.02 ($C_7H_7O_7^-$), m/z^- 242.99 ($C_9H_7O_8^-$) and m/z^- 273.00 ($C_{10}H_9O_9^-$) are also identified as a result of radical reactions. For example, m/z^- 172.99 ($C_6H_5O_6^-$) is the molecular ion peak of 2,5-dioxohexanedioic acid formed by combing two $\dot{C}H_2C(O)CO_2H$ radicals (Rapf et al., 2017). While in the control samples, the observation of m/z^- 175.02 ($C_6H_7O_6^-$), m/z^- 270.97 ($C_9H_3O_{10}^-$), and m/z^- 359.04 ($C_{15}H_{11}O_{12}^-$) suggests PA self-oligomerization and anhydride reactions (Perkins et al., 2016), which are different from the radical reactions at the air-liquid interface (see details in Table S4b). Oxygen could form peroxyl radicals RO_2 (e.g., $CH_3(O_2\cdot)COCOOH$) in the presence of UV light (Bernard et al., 2016). RO_2·radicals have a strong oxidative capacity which could oxidize species with aldehyde functional groups (e.g., m/z^- 225.00 $C_9H_5O_7^-$) to species with carboxyl functional groups (e.g., m/z^- 241.00 $C_9H_5O_8^-$) at the air-liquid interface (Voss et al., 2006; Zhou et al., 2014). The observation that more products are formed by radical reactions indicates that radical reaction is favored to occur at the air-liquid interface.

3.2.2. Non-radical reactions

Besides radical reactions, non-radical reactions (e.g., decarboxylation reaction, anhydride reaction and esterification) contribute to SOA formation in the atmosphere (Altieri et al., 2008; Duplissy et al., 2008; Gao et al., 2004). However, they are highly affected by the species concentration, liquid acidity, and water solubility in the aqueous phase. These factors could vary in the air-liquid interface and the bulk liquid phase resulting in dissimilar products.

Compounds from anhydride reactions (e.g., m/z^- 244.98 $C_9H_9O_8^-$, m/z^- 338.99 $C_{13}H_7O_{11}^-$) and esterification (e.g., m/z^- 157.01 $C_6H_5O_5^-$, m/z^- 198.98 $C_7H_3O_7^-$, m/z^- 268.98 $C_{10}H_5O_9^-$) are favored to form at the air-liquid interface driven likely by photolysis (Fig. 3). m/z^- 244.98 ($C_9H_9O_8^-$) could be formed by undergoing anhydride reactions by parapyruvic acid (m/z^- 175.02, $C_6H_7O_6^-$) and pyruvic acid (m/z^- 87.01, $C_3H_3O_3^-$) (Fig. 3). m/z^- 338.99 ($C_{13}H_7O_{11}^-$) is also generated by the anhydride reaction from m/z^-

268.98 ($C_{10}H_5O_9^-$) and m/z$^-$ 87.01 ($C_3H_3O_3^-$). Due to the presence of carboxyl and hydroxyl functional groups, esterification could easily take place in acidic solutions to form new compounds (e.g., m/z$^-$ 157.01$C_6H_5O_5^-$, m/z 198.98$C_7H_3O_7^-$, m/z$^-$ 268.98$C_{10}H_5O_9^-$). For instance, m/z$^-$ 157.01 ($C_6H_5O_5^-$) was formed from m/z$^-$ 175.02 ($C_6H_7O_6^-$), the latter is a product from an intra-molecular dehydration between a carboxyl functional group and a hydroxyl functional group (Rapf et al., 2017). Non-radical reactions could also be observed in the bulk liquid phase samples, and they are suspected mainly due to drying during sample preparation. For example, m/z$^-$ 157.01 ($C_6H_5O_5^-$) and m/z$^-$ 159.03 ($C_6H_7O_5^-$) are generated by esterification in the bulk liquid phase. Peaks of m/z$^-$ 291.00 ($C_9H_7O_{11}^-$) and m/z$^-$ 317.06 ($C_{12}H_{13}O_{10}^-$) are formed by anhydride reactions. Peaks of m/z$^-$ 113.02 ($C_5H_5O_3^-$) and m/z$^-$ 303.07 ($C_{12}H_{15}O_9^-$) are produced by decarboxylation reactions. The pH of the bulk aqueous samples ranges between 1.79 to 1.82, and mostly acidic (Table S8). Our results support the notion that non-radical reactions could contribute to SOA formation in the atmosphere, especially in the presence of H^+ (Carlton et al., 2007).

3.3. Selected peak spectral PCA

To better understand the differences between the air-liquid interface samples and the bulk liquid phase samples, selected peak spectral PCA is used (Fig. 5 and Fig. S7). 97 representative peaks are chosen in the negative mode. In the 3D score plot in the negative mode (Fig. S7), PC1, PC2, and PC3 explain 59.1%, 24.8%, 5.6% of the variance, respectively, or 89.5% of the variance together. 2D score plots are shown in Fig. 5 and Fig. S7. Different samples are separated in the score plots and distinctions explained by corresponding loading plots (Fig. 5b and c). Spectral PCA in the positive mode is also shown in Figs. S6 and S8.

The two scenarios, the air-liquid interface and the bulk liquid phase, are well distinguished by PC1 from the score plots (Fig. 5a). Positive PC1 could explain the mechanisms at the air-liquid interface. From the loading plots (Fig. 5b and c), compounds like m/z$^-$ 145.01 $C_5H_5O_5^-$, m/z$^-$ 146.99 $C_4H_3O_6^-$, m/z$^-$ 172.99 $C_6H_5O_6^-$, m/z$^-$ 198.98 $C_7H_3O_7^-$, m/z$^-$ 200.95 $C_6HO_8^-$, m/z$^-$ 268.98 $C_{10}H_5O_9^-$, m/z$^-$ 270.97 $C_9H_3O_{10}^-$, m/z$^-$ 338.99 $C_{13}H_7O_{11}^-$, m/z$^-$ 341.03 $C_{13}H_9O_{11}^-$, m/z$^-$ 357.01 $C_{13}H_9O_{12}^-$, m/z$^-$ 359.04 $C_{15}H_{11}O_{12}^-$ are found to contribute significantly to positive PC1. Different mechanisms play an important role in forming these products. For examples, radical reactions are activated at the air-liquid interface contributing to the SOA formation. Two C_2HO_3·radicals could combine to form a compound with m/z$^-$ of 145.01 ($C_5H_5O_5^-$) at the interface. m/z$^-$ 146.99 ($C_4H_3O_6^-$) could be formed via radical reaction between $C_2H_3O_3$·and C_2HO_3. Besides, decarboxylation of the carboxyl functional group is found at the air-liquid interface, representing a few species (e.g., m/z$^-$ 225.00 $C_9H_5O_7^-$, m/z$^-$ 226.99 $C_8H_3O_8^-$ in the positive PC1 loadings) generated by thermal decarboxylation from a larger oligomer produced by radical – radical recombination (Rapf et al., 2017). Moreover, in the presence of hydroxyl and carboxyl functional groups, esterification is identified to participate in SOA formation. For instance, m/z$^-$ 268.98 ($C_{10}H_5O_9^-$) and m/z$^-$ 341.03 ($C_{13}H_9O_{11}^-$) (Fig. 5b) are likely produced via esterification. The PCA results further support SIMS spectral analysis discussed above.

The bulk liquid phase reaction products are located in negative PC1, which could be explained by negative PC1 loadings (Fig. 5b). In the bulk liquid phase, compounds with m/z$^-$ of 157.01 ($C_6H_5O_5^-$), 159.03 ($C_6H_7O_5^-$), 175.02 ($C_6H_5O_6^-$), 177.03 ($C_6H_9O_6^-$), 203.02 ($C_7H_7O_7^-$), 317.06 ($C_{12}H_{13}O_{10}^-$), 331.07 ($C_{13}H_{15}O_{10}^-$), and 347.06 ($C_{13}H_{15}O_{11}^-$) are observed. m/z$^-$ 175.02 ($C_6H_5O_6^-$), m/z$^-$ 177.03 ($C_6H_9O_6^-$), and m/z$^-$ 203.02 ($C_7H_7O_7^-$) are formed by radical reactions (Rxn. 1) shown in Fig. 1. The other products are formed by non-radical reactions in the bulk liquid phase compared to the air-liquid interface. As mentioned above, m/z$^-$ 157.01 ($C_6H_5O_5^-$) and m/z$^-$ 159.03 ($C_6H_7O_5^-$) are likely produced via intermolecular esterification by losing a water molecule. Peaks of m/z$^-$ 317.06 ($C_{12}H_{13}O_{10}^-$) and m/z- 347.06 ($C_{13}H_{15}O_{11}^-$) account for products of anhydride reaction. In addition, m/z$^-$ 303.07 ($C_{12}H_{15}O_9^-$) is produced by decarboxylation of m/z$^-$ 347.06 ($C_{13}H_{15}O_{11}^-$).

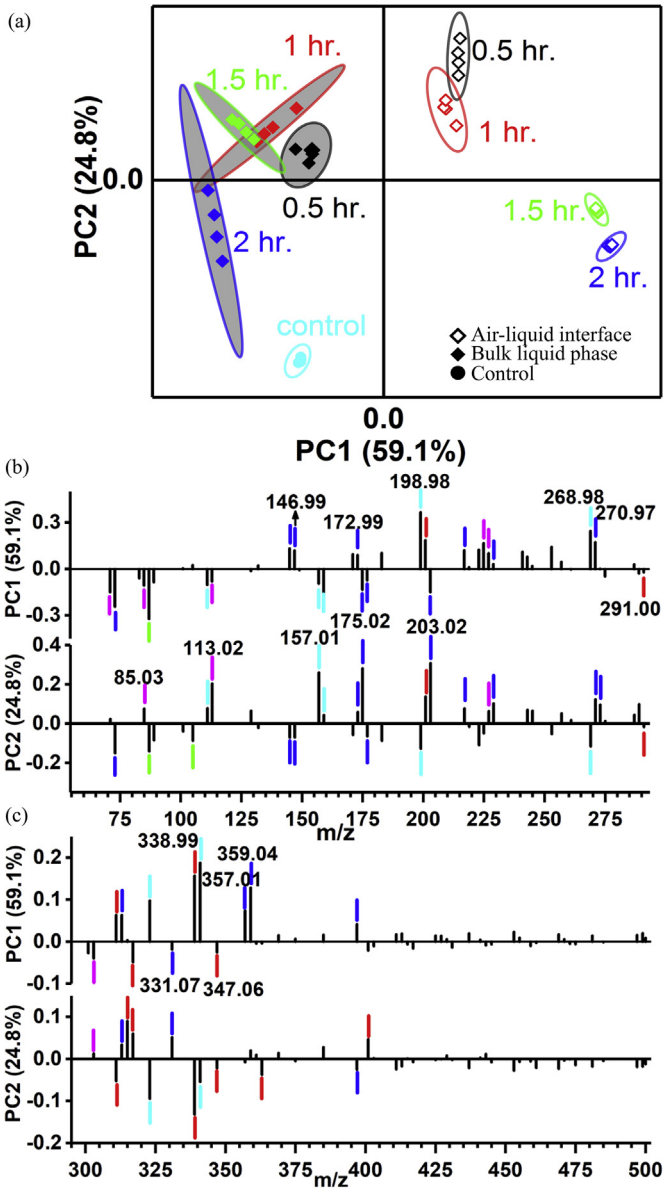


Fig. 5. Selected peak spectral PCA of the air-liquid interface and the bulk liquid phase in the negative mode: a) PC1 vs. PC2 score plot; (b) PC1 and PC2 loading plots in m/z$^-$ 50–300; and c) in m/z$^-$ 300–500. Color bars in loading plots represent different reaction mechanisms. Circles indicate 95% confidence in the PCA results. The bulk liquid phase circles are shaded in gray except the control sample. Green indicates reactant molecular peak, pink decarboxylation, blue radical reaction, red anhydride reaction, and cyan esterification. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In order to understand the effect of photochemistry time on the products formed at the air-liquid interface and the bulk liquid phase, four time-dependent photochemical aging samples are studied. The time effect on the air-liquid interface is clearly illustrated among samples ranging from 0.5 h to 2 h in Fig. 5a. Shorter time samples reside in the PC1 positive and PC2 positive quadrant, and the longer time samples in the PC1 positive and PC2 negative quadrant. Therefore, PC2 loadings distinguish the products at the air-liquid interface dependence on reaction time. According to the loading plots, the shorter reaction

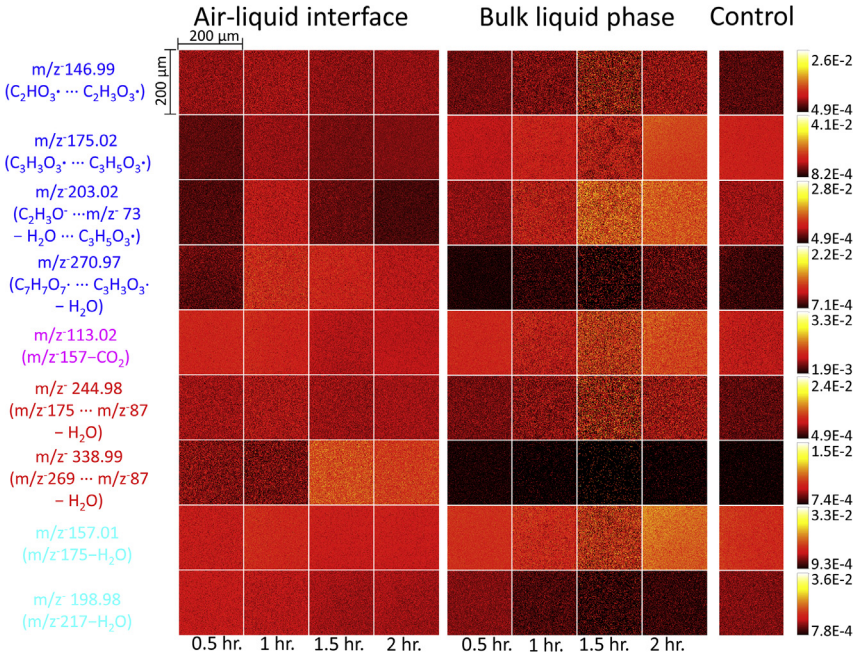


Fig. 6. Comparison of 2D images of characteristic peaks normalized to the total ion intensities in the negative ion mode. Brighter color represents higher counts and darker colour lower counts. The colored text on the left column represents different reaction mechanisms, blue indicates radical reaction, pink decarboxylation, red anhydride reaction, and cyan esterification. In SIMS 2D images, brighter color indicates higher relative counts. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

time facilitates radical reactions mainly involving $CH_3\dot{C}(OH)CO_2H$ and $\dot{C}H_2C(O)CO_2H$ radicals which are formed from the PA molecules. The representative peaks of the products included m/z$^-$ 172.99 ($C_6H_5O_6^-$), m/z$^-$ 216.99 ($C_7H_5O_8^-$), m/z$^-$ 229.03 ($C_9H_9O_7^-$) and m/z$^-$ 270.97 ($C_9H_3O_{10}^-$) (the detailed information can be found in Tables S4a and 4b).

During drying, liquid would disappear at around 1.5 h in our laboratory. After 1.5 h of photochemical aging at the air-liquid interface, the species remain consistent as the photochemical aging continues which is depicted in the spectra plot in Fig. 3a. During drying up to 1.5 h, the effect of radical reactions is less significant compared to dehydration reactions occurred (e.g., esterification and anhydride reactions) at the air-liquid interface due to the overall smaller liquid volume compared to the bulk liquid scenario. Representative compounds (e.g., m/z$^-$ 198.98 $C_7H_3O_7^-$, m/z$^-$ 268.98 $C_{10}H_5O_9^-$, m/z$^-$ 323.01 $C_{13}H_7O_{10}^-$, m/z$^-$ 341.03 $C_{13}H_9O_{11}^-$ from the esterification reaction and m/z$^-$ 338.99 $C_{13}H_7O_{11}^-$, m/z$^-$ 311.00 $C_{12}H_7O_{10}^-$ in the anhydride reaction) are observed from samples with less than 1.5 h photolysis. The different products and mechanisms during the combined photolysis and drying process at the air-liquid interface indicate that the effect of water evaporation and photochemistry at the particle aqueous surface of the fog and cloud water would generate oxidation products and they in turn influence the global climate change (Vaida, 2016). Our SIMS surface analysis provides additional evidence of this earlier finding with new insight on the surface heterogeneous reaction product complexity.

Photochemical aging time is one of the main factors discriminating the bulk liquid phase samples. According to the score plots (Fig. 5a and Fig. S7), the shorter time sample, namely 0.5 h in the bulk liquid phase, is located in the negative PC1, positive PC2, and negative PC3 area. The representative products (e.g., m/z$^-$ 85.03 $C_4H_5O_2^-$, m/z$^-$ 113.02 $C_5H_5O_3^-$, m/z$^-$ 157.01 $C_6H_5O_5^-$, m/z$^-$ 175.02 $C_6H_7O_6^-$) are from radical reactions and non-radical reactions such as decarboxylation and esterification, suggesting that m/z$^-$ 85.03 ($C_4H_5O_2^-$) and m/z$^-$ 113.02 ($C_5H_5O_3^-$) would form by reduction of small molecules such as $H_2C_2O_4$ and CO_2. After 0.5 h photochemical aging, many products are observed including m/z$^-$ 159.03 $C_6H_7O_5^-$, m/z$^-$ 177.03 $C_6H_9O_6^-$, as well as HMWCs such as m/z$^-$ 203.02 $C_7H_7O_7^-$, m/z$^-$ 275.00 $C_{10}H_{11}O_9^-$, m/z$^-$ 303.07, m/z$^-$ 317.06 $C_{12}H_{13}O_{10}^-$, and m/z$^-$ 347.06 $C_{13}H_{15}O_{11}^-$. The larger HMWCs are derived by anhydride reactions (e.g., m/z$^-$ 317.06 $C_{12}H_{13}O_{10}^-$, m/z$^-$ 347.06 $C_{13}H_{15}O_{11}^-$). Such results indicate that the initial photolysis of PA may induce the low molecular weight oligomers formation in the bulk liquid phase, which could further react with other compounds via acid-catalysis reactions, for instance, and form HMWCs, as shown in Rxns. 2–4 in Fig. 1. PC2 loadings also suggest the main product differences among samples treated with photolysis of less than 1.5 h and the sample treated with longer photolysis time. Specifically, PC2 negative loading peaks (Fig. 5b and c) such as m/z$^-$ 177.03 $C_6H_9O_6^-$, m/z$^-$ 268.98 $C_{10}H_5O_9^-$, m/z$^-$ 291.00 $C_9H_7O_{11}^-$, m/z$^-$ 323.01 $C_{13}H_7O_{10}^-$, m/z$^-$ 317.06 $C_{12}H_{13}O_{10}^-$, m/z$^-$ 338.99 $C_{13}H_7O_{11}^-$, m/z$^-$ 347.06 $C_{13}H_{15}O_{11}^-$, m/z$^-$ 363.06 $C_{13}H_{15}O_{12}^-$ indicate reaction products become more important after longer photolysis.

Overall our results show that sample preparation approaches and atmospheric processes affect the observed oxidation products. For the air-liquid interface samples, drying takes place simultaneously with photolysis, which leads to concentration enrichment. Consequently, more radical reactions occur at the air-liquid interface. Along with radical reactions, other compounds from non-radical reactions contribute to the SOA formation at the air-liquid interface. As to the bulk liquid phase samples, photochemistry occurs before drying, providing a different reaction environment that leads to distinctive products.

3.4. 2D image and depth profiles

As a powerful imaging tool, ToF-SIMS is often used to provide molecular mapping at the surface of dry samples (Hagenhoff, 2000). A comparison of 2D negative ion images of the air-liquid interface samples and the bulk liquid phase samples are depicted in Fig. 6. The brightness of the image corresponds to the normalized intensity of the peaks. Characteristic peaks, such as m/z$^-$ 113.02 $C_5H_5O_3^-$, m/z$^-$ 146.99 $C_4H_3O_6^-$, m/z$^-$ 157.01 $C_6H_5O_5^-$, m/z$^-$ 175.02 $C_6H_7O_6^-$, m/z$^-$ 198.98 $C_7H_3O_7^-$, m/z$^-$ 203.02 $C_7H_7O_7^-$, m/z$^-$ 244.98 $C_9H_9O_8^-$, m/z$^-$ 270.97 $C_9H_3O_{10}^-$, and m/z$^-$ 338.99 $C_{13}H_7O_{11}^-$, representing different reaction mechanisms are compared. From the 2D images, m/z$^-$ 146.99 $C_4H_3O_6^-$ and m/z$^-$ 270.97 $C_9H_3O_{10}^-$ have brighter images at the air-liquid interface than the bulk liquid phase, while m/z$^-$ 175.02 $C_6H_7O_6^-$ and m/z$^-$ 203.02 $C_7H_7O_7^-$ show the opposite trend. All these products are formed from radical reactions, providing a direct evidence to support the hypothesis that radical reactions exist on both

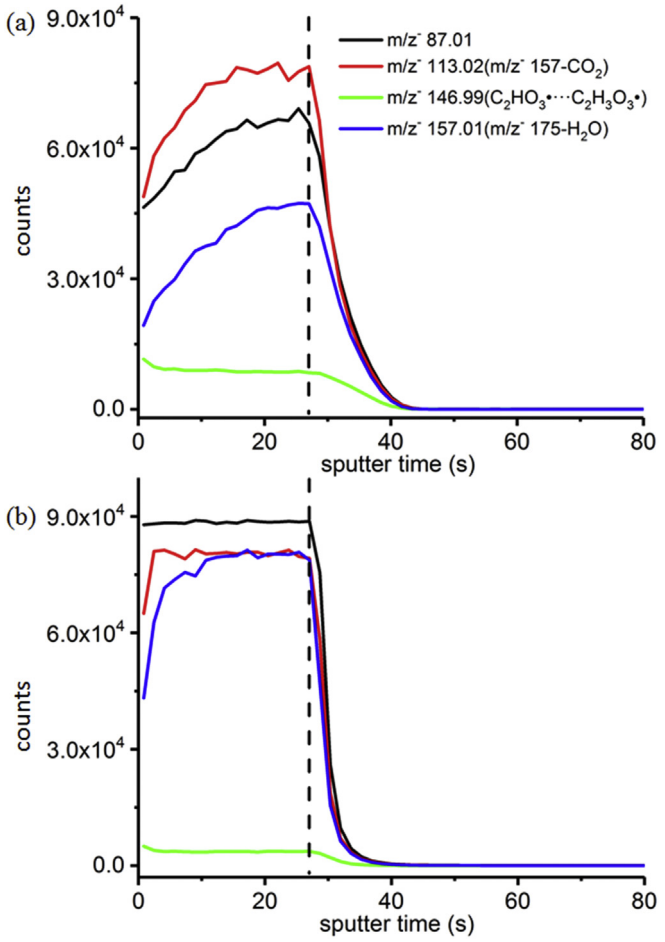


Fig. 7. SIMS depth profiles showing the representative ions at the air-liquid interface (a) and bulk liquid phase (b) composition changes of the 1.5 h samples in the negative mode. The dash line represents the time when the sample on the Si wafer are punched through.

scenarios. A recent paper showed that anhydride reactions also played an important role in forming HMWCs (e.g., m/z$^-$ 244.98 $C_9H_9O_8^-$ and m/z$^-$ 338.99 $C_{13}H_7O_{11}^-$) at the air-liquid interface, which could contribute to the SOA loading in the atmosphere (Vaida, 2016). Esterification and decarboxylation are also found to produce SOAs in the bulk liquid phase (e.g., m/z$^-$ 157.01 $C_6H_5O_5^-$ and m/z$^-$ 198.98 $C_7H_3O_7^-$ from esterification, m/z$^-$ 113.02 $C_5H_5O_3^-$ from decarboxylation) in this work, as reflected in their higher relative intensities. 2D image results directly provide the evidence that different radical reactions occur in two scenarios. At the air-liquid interface, the radical reactions are not only involved in radicals formed by PA and PA*, but also some other radicals (e.g., two C_2HO_3· formed m/z$^-$ 146.99 $C_4H_3O_6^-$). While in the bulk liquid phase, the radical reactions are mainly related to the radicals formed by PA and PA* (e.g., $C_3H_3O_3$· and $C_3H_5O_3$· formed m/z$^-$ 175.02 $C_6H_7O_6^-$). Besides, products generated by radical reactions at the air-liquid interface have larger molecular weight, such as m/z$^-$ 270.97 $C_9H_3O_{10}^-$ ($C_7H_7O_7$· ··· $C_3H_3O_3$· − H_2O). Our results give observations to support the postulation raised in earlier works that water evaporation could lead to an unstable dynamic aqueous environment due to the energy transfer or changes at the air-liquid interface (Sitzmann and Eisenthal, 1989; Zhang et al., 2011).

Fig. 7 and Fig. S9 show the depth profiles of representative ions at the air-liquid interface. The intensities of m/z$^-$ 146.99 ($C_4H_3O_6^-$), m/z$^-$ 198.98 ($C_7H_3O_7^-$) in Fig. S9 and m/z$^-$ 270.97 ($C_9H_3O_{10}^-$) decrease as the depth increased, which indicates that radical reactions are facilitated at the surface. While the intensity of m/z$^-$ 113.02 ($C_5H_5O_3^-$) from decarboxylation increases as the depth increases, implying that the decarboxylation favors to form in the inner layers under the surface. In the bulk liquid phase dry samples, the trend is different from that in the air-liquid interface samples. The intensities of the HWMCs (e.g. m/z$^-$ 270.97 $C_9H_3O_{10}^-$, m/z$^-$ 338.99 $C_{13}H_7O_{11}^-$) in the bulk liquid phase are lower than those at the air-liquid interface. Interestingly, the intensities of the products in the bulk liquid phase samples are steady as the depth increases, which indicates that the bulk liquid phase samples are more homogeneous. The intensities of radical reaction products (m/z$^-$ 146.99 ($C_4H_3O_6^-$) and m/z$^-$ 270.97 ($C_9H_3O_{10}^-$)) do not change significantly when increasing the depth into the sample, implying that the radical reactions occur evenly in the bulk liquid phase. Also the non-radical reactions are found to take place evenly in the bulk liquid phase samples, reflected by the stable ion intensities along the depth profiling.

4. Conclusions

We investigate two simulated scenarios of the atmosphere photochemistry involving PA in liquid water: the air-liquid interface and the bulk liquid phase under photolysis. Different kinds of products are observed from the dry SIMS mass spectral analysis of the two scenarios, demonstrating that the sample preparation methods and steps are critical for understanding oxidation products and reaction pathways. From the spectral PCA results, the effects of UV aging time and sample preparation methods are elucidated. At the air-liquid interface, the combined effects of photochemistry and water evaporation result in more radical reactions and formation of HMWCs. While in the bulk aqueous phase, non-radical reactions are mainly observed. The short time UV aging at the air-liquid interface favors radical reactions related to radicals formed by PA molecules and longer-time reactions facilitate the radical reactions engaging other radicals. While for the bulk liquid phase samples, the shorter-time reaction favors small oligomer product formation. As the UV aging time increases, the compounds would grow to larger masses. SIMS 2D image and depth profile results provide direct molecular information of the samples. At the air-liquid interface, water evaporation and photochemistry cause unstable dynamic environment due to energy transfer at the surface, resulting in more radical reaction products at the interfaces. While the bulk liquid phase offers a relatively steady environment or no vigorous water evaporation compared to the air-liquid interface. Such findings suggest that more radical reactions could happen and HMWCs would form quickly as water evaporates from the aerosol surface due to temperature elevation, contributing to aqSOA formation.

It is worth noting that results discussed in this work are acquired using SIMS analysis of dry samples. Interferences from the drying effect are unavoidable. In the atmosphere, PA photochemistry occurs in a complex environment, representing a multiphase and interfacial challenge. Moreover, the water structure due to hydrogen bonding in an aqueous solution and its response to the presence of hydrophilic and hydrophobic solutes determine the surface composition and surface tension (Brini et al., 2017; Garrett et al., 2006), which can play an important role in the chemical dynamics of the air-liquid interface and liquid surface. This information cannot be obtained by dry sample analysis, therefore it is necessary to study the air-liquid interface and the aqueous solution surface directly using in situ liquid SIMS (Yang et al., 2011a, 2011b) in the next step.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.atmosenv.2018.07.061.

References

Altieri, K.E., Carlton, A.G., Lim, H.J., Turpin, B.J., Seitzinger, S.P., 2006. Evidence for oligomer formation in clouds: reactions of isoprene oxidation products. Environ. Sci. Technol. 40, 4956–4960.

Altieri, K.E., Seitzinger, S.P., Carlton, A.G., Turpin, B.J., Klein, G.C., Marshall, A.G., 2008. Oligomers formed through in-cloud methylglyoxal reactions: chemical composition, properties, and mechanisms investigated by ultra-high resolution FT-ICR mass spectrometry. Atmos. Environ. 42, 1476–1490.

Andreae, M.O., Talbot, R.W., Li, S.M., 1987. Atmospheric measurements of pyruvic and formic-acid. J. Geophys. Res.: Atmos. 92, 6635–6641.

Arnett, J.F., Larson, D.B., Mcglynn, S.P., 1974. Absorption and emission spectroscopy of pyruvic acids and pyruvate esters. J. Am. Chem. Soc. 5, 7599–7603.

Baergen, A.M., Donaldson, D.J., 2013. Photochemical renoxification of nitric acid on real urban grime. Environ. Sci. Technol. 47, 815–820.

Benninghoven, A., 1994. Chemical analysis of inorganic and organic surfaces and thin films by static time-of-flight secondary ion mass spectrometry (TOF-SIMS). Angew. Chem. Int. Ed. 33, 1023–1043.

Bernard, F., Ciuraru, R., Boreave, A., George, C., 2016. Photosensitized formation of secondary organic aerosols above the air/water interface. Environ. Sci. Technol. 50, 8678–8686.

Boris, A.J., Desyaterik, Y., Collett, J.L., 2014. How do components of real cloud water affect aqueous pyruvate oxidation? Atmos. Res. 143, 95–106.

Brüggemann, M., Hayeck, N., Bonnineau, C., Pesce, S., Alpert, P.A., Perrier, S., Zuth, C., Hoffmann, T., Chen, J.M., George, C., 2017. Interfacial photochemistry of biogenic surfactants: a major source of abiotic volatile organic compounds. Faraday Discuss 200, 59–74.

Brini, E., Fennell, C.J., Fernandez-Serra, M., Hribar-Lee, B., Lukšič, M., Dill, K.A., 2017. How Water's properties are encoded in its molecular structure and energies. Chem. Rev. 117, 12385–12414.

Buch, V., Milet, A., Vacha, R., Jungwirth, P., Devlin, J.P., 2007. Water surface is acidic. Proc. Natl. Acad. Sci. Unit. States Am. 104, 7342–7347.

Carlton, A.G., Turpin, B.J., Lim, H.-J., Altieri, K.E., Seitzinger, S., 2006. Link between isoprene and secondary organic aerosol (SOA): pyruvic acid oxidation yields low volatility organic acids in clouds. Geophys. Res. Lett. 33.

Carlton, A.G., Wiedinmyer, C., Kroll, J.H., 2009. A review of Secondary Organic Aerosol (SOA) formation from isoprene. Atmos. Chem. Phys. 9, 4987–5005.

Carlton, A.M.,J., Turpin, B., Altieri, K., Seitzinger, S., Reff, A., Lim, H.-J., Ervens, B., 2007. Atmospheric oxalic acid and SOA production from glyoxal: results of aqueous photooxidation experiments. Atmos. Environ. 41, 7588–7602.

Chan, Q.H.S., Zolensky, M.E., Kebukawa, Y., Fries, M., Ito, M., Steele, A., Rahman, Z., Nakato, A., Kilcoyne, A.L.D., Suga, H., Takahashi, Y., Takeichi, Y., Mase, K., 2018. Organic matter in extraterrestrial water-bearing salt crystals. Sci. Adv. 4.

Chang, X.P., Fang, Q., Cui, G., 2014. Mechanistic photodecarboxylation of pyruvic acid: excited-state proton transfer and three-state intersection. J. Chem. Phys. 141, 154311.

Ciuraru, R., Fine, L., van Pinxteren, M., D'Anna, B., Herrmann, H., George, C., 2015. Photosensitized production of functionalized and unsaturated organic compounds at the air-sea interface. Sci. Rep. 5, 12741.

Ding, Y., Zhou, Y., Yao, J., Szymanski, C., Fredrickson, J., Shi, L., Cao, B., Zhu, Z., Yu, X.Y., 2016. In situ molecular imaging of the biofilm and its matrix. Anal. Chem. 88, 11244–11252.

Donaldson, D.J., Vaida, V., 2006. The influence of organic films at the air-aqueous boundary on atmospheric processes. Chem. Rev. 106, 1445–1461.

Dupart, Y., King, S.M., Nekat, B., Nowak, A., Wiedensohler, A., Herrmann, H., David, G., Thomas, B., Miffre, A., Rairoux, P., D'Anna, B., George, C., 2012. Mineral dust photochemistry induces nucleation events in the presence of SO2. Proc. Natl. Acad. Sci. Unit. States Am. 109, 20842–20847.

Duplissy, J., Gysel, M., Alfarra, M.R., Dommen, J., Metzger, A., Prevot, A.S.H., Weingartner, E., Laaksonen, A., Raatikainen, T., Good, N., Turner, S.F., McFiggans, G., Baltensperger, U., 2008. Cloud forming potential of secondary organic aerosol under near atmospheric conditions. Geophys. Res. Lett. 35.

Eisenthal, K.B., 1993. Liquid interfaces. Acc. Chem. Res. 26, 636–643.

Enami, S., Hoffmann, M.R., Colussi, A.J., 2015. Stepwise oxidation of aqueous dicarboxylic acids by gas-phase OH radicals. J. Phys. Chem. Lett. 6, 527–534.

Ervens, B., Turpin, B.J., Weber, R.J., 2011. Secondary organic aerosol formation in cloud droplets and aqueous particles (aqSOA): a review of laboratory, field and model studies. Atmos. Chem. Phys. 11, 11069–11102.

Eugene, A.J., Guzman, M.I., 2017. Reactivity of ketyl and acetyl radicals from direct solar actinic photolysis of aqueous pyruvic acid. J. Phys. Chem. 121, 2924–2935.

Fearn, S., 2015. Characterisation of biological material with ToF-SIMS: a review. Mater. Sci. Technol. 31, 148–161.

Fu, H., Ciuraru, R., Dupart, Y., Passananti, M., Tinel, L., Rossignol, S., Perrier, S., Donaldson, D.J., Chen, J., George, C., 2015. Photosensitized production of atmospherically reactive organic compounds at the air/aqueous interface. J. Am. Chem. Soc. 137, 8348–8351.

Gao, S., Ng, Nga L., Keywood, Melita, Varutbangkul, Varuntida, Bahreini, Roya, Nenes, Athanasios, He, Jiwen, Yoo, Kee Y., Beauchamp, J.L., Hodyss, Robert P., 2004. Particle phase acidity and oligomer formation in secondary organic aerosol. Environ. Sci. Technol. 38, 6582–6589.

Garrett, B.C., Schenter, G.K., Morita, A., 2006. Molecular simulations of the transport of molecules across the liquid/vapor interface of water. Chem. Rev. 106, 1355–1374.

George, C., Ammann, M., D'Anna, B., Donaldson, D.J., Nizkorodov, S.A., 2015. Heterogeneous photochemistry in the atmosphere. Chem. Rev. 115, 4218–4258.

Griffith, E.C., Carpenter, B.K., Shoemaker, R.K., Vaida, V., 2013. Photochemistry of aqueous pyruvic acid. Proc. Natl. Acad. Sci. Unit. States Am. 110, 11714–11719.

Griffith, E.C., Rapf, R.J., Shoemaker, R.K., Carpenter, B.K., Vaida, V., 2014. Photoinitiated synthesis of self-assembled vesicles. J. Am. Chem. Soc. 136, 3784–3787.

Guzman, M.I., Colussi, A.J., Hoffmann, M.R., 2006. Photoinduced oligomerization of aqueous pyruvic acid. J. Phys. Chem. 110, 3619–3626.

Hagenhoff, B., 2000. High resolution surface analysis by TOF-SIMS. Mikrochim. Acta 132, 259–271.

Horowitz, A., Meller, R., Moortgat, G.K., 2001. The UV-VIS absorption cross sections of the alpha-dicarbonyl compounds: pyruvic acid, biacetyl and glyoxal. J. Photochem. Photobiol., A 146, 19–27.

Huang, D., Xiu, G., Li, M., Hua, X., Long, Y., 2017. Surface components of PM2.5 during clear and hazy days in Shanghai by ToF-SIMS. Atmos. Environ. 148, 175–181.

Jensen, D.S., Kanyal, S.S., Madaan, N., Hancock, J.M., Dadson, A.E., Vail, M.A., Vanfleet, R., Shutthanandan, V., Zhu, Z., Engelhard, M.H., Linford, M.R., 2013. Multi-instrument characterization of the surfaces and materials in microfabricated, carbon nanotube-templated thin layer chromatography plates. An analogy to 'The Blind Men and the Elephant'. Surf. Interface Anal. 45, 1273–1282.

Jungwirth, P., Tobias, D.J., 2006. Specific ion effects at the air/water interface. Chem. Rev. 106, 1259–1281.

Kanyal, S.S., Jensen, D.S., Zhu, Z., Linford, M.R., 2015. Silicon (100)/SiO2 by ToF-SIMS. Surf. Sci. Spectra 22, 1–6.

Krieger, U.K., Marcolli, C., Reid, J.P., 2012. Exploring the complexity of aerosol particle properties and processes using single particle techniques. Chem. Soc. Rev. 41, 6631–6662.

Lanni, E.J., Masyuko, R.N., Driscoll, C.M., Dunham, S.J.B., Shrout, J.D., Bohn, P.W., Sweedler, J.V., 2014. Correlated imaging with C60-SIMS and confocal Raman microscopy: visualization of cell-scale molecular distributions in bacterial biofilms. Anal. Chem. 86, 10885–10891.

Larsen, M.C., Vaida, V., 2012. Near infrared photochemistry of pyruvic acid in aqueous solution. J. Phys. Chem. 116, 5840–5846.

Leermakers, P.A., Vesley, G.F., 1963. The photochemistry of α-keto acids and α-keto esters. I. Photolysis of pyruvic acid and benzoylformic acid. J. Am. Chem. Soc. 85, 3776–3779.

Lim, Y.B., Tan, Y., Perri, M.J., Seitzinger, S.P., Turpin, B.J., 2010. Aqueous chemistry and its role in secondary organic aerosol (SOA) formation. Atmos. Chem. Phys. 10, 10521–10539.

Lopalco, A., Dalwadi, G., Niu, S., Schowen, R.L., Douglas, J., Stella, V.J., 2016. Mechanism of decarboxylation of pyruvic acid in the presence of hydrogen peroxide. J. Pharmacol. Sci. 105, 705–713.

Mahoney, C.M., Gillen, G., Fahey, A.J., 2006. Characterization of gunpowder samples using time-of-flight secondary ion mass spectrometry (TOF-SIMS). Forensic Sci. Int. 158, 39–51.

Mellouki, A., Mu, Y., 2003. On the atmospheric degradation of pyruvic acid in the gas phase. J. Photochem. Photobiol., A 157, 295–300.

Perkins, R.J., Shoemaker, R.K., Carpenter, B.K., Vaida, V., 2016. Chemical equilibria and kinetics in aqueous solutions of zymonic acid. J. Phys. Chem. 120, 10096–10107.

Petersen, M.K., Iyengar, S.S., Day, T.J.F., Voth, G.A., 2004. The hydrated proton at the water liquid/vapor interface. J. Phys. Chem. B 108, 14804–14806.

Petersen, P.B., Saykally, R.J., 2006. ON the NATURE of ions at the liquid water surface. Annu. Rev. Phys. Chem. 57, 333–364.

Peterson, R.E., Tyler, B.J., 2002. Analysis of organic and inorganic species on the surface of atmospheric aerosol using time-of-flight secondary ion mass spectrometry (TOF-SIMS). Atmos. Environ. 36, 6041–6049.

Peterson, R.E., Tyler, B.J., 2003. Surface composition of atmospheric aerosol: individual particle characterization by TOF-SIMS. Appl. Surf. Sci. 203, 751–756.

Plath, K.L., Takahashi, K., Skodje, R.T., Vaida, V., 2009. Fundamental and overtone vibrational spectra of gas-phase pyruvic acid. J. Phys. Chem. 113, 7294–7303.

Pruyne, J.G., Lee, M.-T., Fábri, C., Beloqui Redondo, A., Kleibert, A., Ammann, M., Brown, M.A., Krisch, M.J., 2014. Liquid–vapor interface of formic acid solutions in salt water: a comparison of macroscopic surface tension and microscopic in situ X-ray photoelectron spectroscopy measurements. J. Phys. Chem. C 118, 29350–29360.

Rapf, R.J., Perkins, R.J., Carpenter, B.K., Vaida, V., 2017. Mechanistic Description of Photochemical Oligomer Formation from Aqueous Pyruvic Acid. J. Phys. Chem. A 122, 4272–4282.

Reed Harris, A.E., Doussin, J.F., Carpenter, B.K., Vaida, V., 2016. Gas-phase photolysis of pyruvic acid: the effect of pressure on reaction rates and products. J. Phys. Chem. 120, 10123–10133.

Reed Harris, A.E., Ervens, B., Shoemaker, R.K., Kroll, J.A., Rapf, R.J., Griffith, E.C., Monod, A., Vaida, V., 2014. Photochemical kinetics of pyruvic acid in aqueous solution. J. Phys. Chem. 118, 8505–8516.

Renard, P., Reed Harris, A.E., Rapf, R.J., Ravier, S., Demelas, C., Coulomb, B., Quivet, E., Vaida, V., Monod, A., 2014. Aqueous phase oligomerization of methyl vinyl ketone by atmospheric radical reactions. J. Phys. Chem. C 118, 29421–29430.

Robinson, M.A., Graham, D.J., Castner, D.G., 2012. ToF-sims depth profiling of cells: z-

correction, 3D imaging, and sputter rate of individual NIH/3T3 fibroblasts. Anal. Chem. 84, 4880–4885.
Rossignol, S., Tinel, L., Bianco, A., Passananti, M., Brigante, M., Donaldson, D.J., George, C., 2016. Atmospheric photochemistry at a fatty acid-coated air-water interface. Science 353, 699–702.
Schöne, L., Herrmann, H., 2014. Kinetic measurements of the reactivity of hydrogen peroxide and ozone towards small atmospherically relevant aldehydes, ketones and organic acids in aqueous solutions. Atmos. Chem. Phys. 14, 4503–4514.
Shiraiwa, M., Pfrang, C., Pöschl, U., 2010. Kinetic multi-layer model of aerosol surface and bulk chemistry (KM-SUB): the influence of interfacial transport and bulk diffusion on the oxidation of oleic acid by ozone. Atmos. Chem. Phys. 10, 3673–3691.
Sitzmann, E.V., Eisenthal, K.B., 1989. Dynamics of intermolecular electronic energy transfer at an air/liquid interface. J. Chem. Phys. 90, 2831–2832.
Song, J., Jensen, D.S., Hutchison, D.N., Turner, B., Wood, T., Dadson, A., Vail, M.A., Linford, M.R., Vanfleet, R.R., Davis, R.C., 2011. Carbon-Nanotube-Templated microfabrication of porous silicon-carbon materials with application to chemical separations. Adv. Funct. Mater. 21, 1132–1139.
Sui, X., Zhou, Y., Zhang, F., Zhang, Y., Chen, J., Zhu, Z., Yu, X.-Y., 2017. Tof-sims characterization of glyoxal surface oxidation products by hydrogen peroxide: A comparison between dry and liquid samples. Surf. Interface Anal. 1–12. https://doi.org/10.1002/sia.6334.
Sui, X., Zhou, Y., Zhang, F., Zhang, Y., Chen, J., Zhu, Z., Yu, X.-Y., 2018. ToF-SIMS characterization of glyoxal surface oxidation products by hydrogen peroxide: a comparison between dry and liquid samples. Surf. Interface Anal. Advanced on line. https://doi.org/10.1002/sia.6334.
Takahashi, K., Plath, K.L., Skodje, R.T., Vaida, V., 2008. Dynamics of vibrational overtone excited pyruvic acid in the gas phase: line broadening through hydrogen-atom chattering. J. Phys. Chem. 112, 7321–7331.
Talbot, R.W., Andreae, M.O., Berresheim, H., Jacob, D.J., Beecher, K.M., 1990. Sources and sinks of formic, acetic, and pyruvic acids over central amazonia .2. Wet season. J. Geophys. Res.: Atmos. 95, 16799–16811.
Tan, Y., Lim, Y.B., Altieri, K.E., Seitzinger, S.P., Turpin, B.J., 2012. Mechanisms leading to oligomers and SOA through aqueous photooxidation: insights from OH radical oxidation of acetic acid and methylglyoxal. Atmos. Chem. Phys. 12, 801–813.
Tervahattu, H., 2002. Identification of an organic coating on marine aerosol particles by TOF-SIMS. J. Geophys. Res. 107.
Vaida, V., 2011. Perspective: water cluster mediated atmospheric chemistry. J. Chem. Phys. 135, 020901.
Vaida, V., 2016. ATMOSPHERIC CHEMISTRY. Atmospheric radical chemistry revisited. Science 353, 650.
Veres, P., Roberts, J.M., Burling, I.R., Warneke, C., de Gouw, J., Yokelson, R.J., 2010. Measurements of gas-phase inorganic and organic acids from biomass fires by negative-ion proton-transfer chemical-ionization mass spectrometry. J. Geophys. Res. 115.
Vesley, G.F., Leermakers, P.A., 1964. The photochemistry of α-keto acids and α-keto esters. Iii. Photolysis of pyruvic acid in the vapor phase. J. Phys. Chem. 68, 2364–2366.
Voss, L.F., Hadad, C.M., Allen, H.C., 2006. Competition between atmospherically relevant fatty acid monolayers at the air/water interface. J. Phys. Chem. B 110, 19487–19490.
Warneck, P., 2005. Multi-phase chemistry of c2 and c3 organic compounds in the marine atmosphere. J. Atmos. Chem. 51, 119–159.
Yang, L., Yu, X.-Y., Zhu, Z., Iedema, M.J., Cowin, J.P., 2011a. Probing liquid surfaces under vacuum using SEM and ToF-SIMS. Lab a Chip 11, 2481.
Yang, L., Yu, X.-Y., Zhu, Z., Thevuthasan, T., Cowin, J.P., 2011b. Making a hybrid microfluidic platform compatible for in situ imaging by vacuum-based techniques. J. Vac. Sci. Technol., A 29, 061101.
Yu, B.-Y., Lin, W.-C., Wang, W.-B., Iida, S.-i., Chen, S.-Z., Liu, C.-Y., Kuo, C.-H., Lee, S.-H., Kao, W.-L., Yen, G.-J., You, Y.-W., Liu, C.-P., Jou, J.-H., Shyue, J.-J., 2010. Effect of fabrication parameters on three-dimensional nanostructures of bulk heterojunctions imaged by high-resolution scanning ToF-SIMS. ACS Nano 4, 833–840.
Zhang, Z., Li, H., Liu, H., Ni, R., Li, J., Deng, L., Lu, D., Cheng, X., Duan, P., Li, W., 2016. A preliminary analysis of the surface chemistry of atmospheric aerosol particles in a typical urban area of Beijing. J. Environ. Sci. 47, 71–81.
Zhang, Z., Piatkowski, L., Bakker, H.J., Bonn, M., 2011. Ultrafast vibrational energy transfer at the water/air interface revealed by two-dimensional surface vibrational spectroscopy. Nat. Chem. 3, 888.
Zhou, S., Gonzalez, L., Leithead, A., Finewax, Z., Thalman, R., Vlasenko, A., Vagle, S., Miller, L.A., Li, S.M., Bureekul, S., Furutani, H., Uematsu, M., Volkamer, R., Abbatt, J., 2014. Formation of gas-phase carbonyls from heterogeneous oxidation of polyunsaturated fatty acids at the air–water interface and of the sea surface microlayer. Atmos. Chem. Phys. 14, 1371–1384.
Zhu, Y.-J., Olson, N., Beebe, T.P., 2001. Surface chemical characterization of 2.5-μm particulates (PM2.5) from air pollution in salt lake city using TOF-SIMS, XPS, and FTIR. Environ. Sci. Technol. 35, 3113–3121.